

Lecture 7 More On The Bayes' Theorem

BIO210 Biostatistics

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Fall, 2026

School of Life Sciences

Southern University of Science and Technology

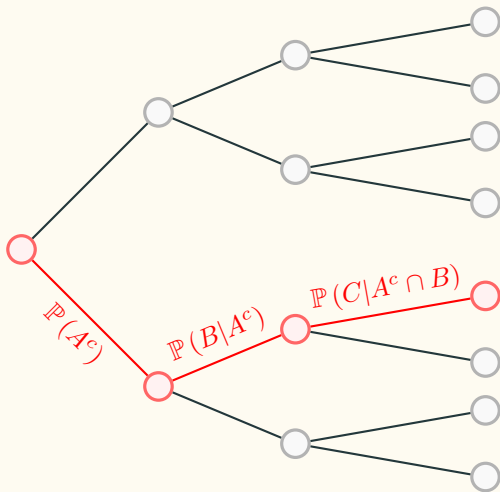


南方科技大学生命科学学院
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Conditional Probability

The Multiplication Rule

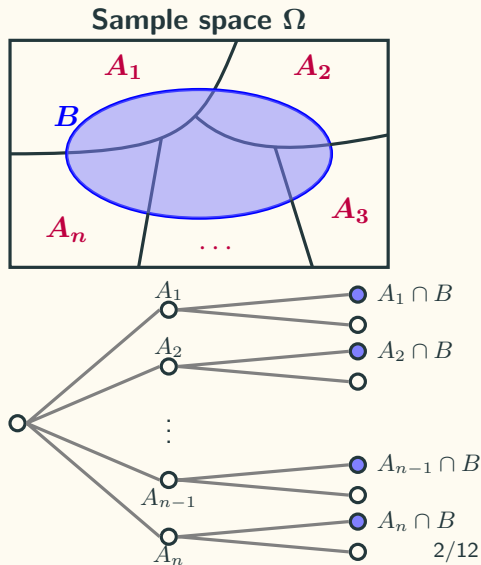
$$\begin{aligned}\mathbb{P}(\cap_{i=1}^n A_i) &= \mathbb{P}(A_1) \cdot \\ &\quad \mathbb{P}(A_2|A_1) \cdot \\ &\quad \mathbb{P}(A_3|A_1 \cap A_2) \cdot \\ &\quad \mathbb{P}(A_4|A_1 \cap A_2 \cap A_3) \cdot \\ &\quad \dots \\ &\quad \mathbb{P}(A_n|\cap_{i=1}^{n-1} A_i)\end{aligned}$$



Conditional Probability

The Total Probability Rule

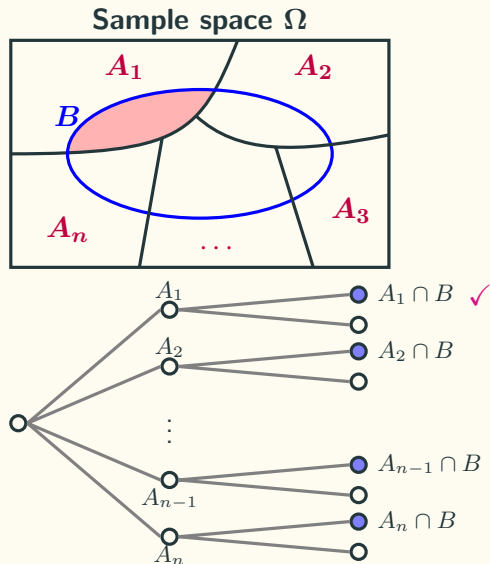
$$\begin{aligned}\mathbb{P}(B) &= \mathbb{P}[(A_1 \cap B) \cup (A_2 \cap B) \cup \dots \cup (A_n \cap B)] \\ &= \mathbb{P}(A_1 \cap B) + \mathbb{P}(A_2 \cap B) + \dots + \mathbb{P}(A_n \cap B) \\ &= \sum_{i=1}^n \mathbb{P}(A_i) \cdot \mathbb{P}(B|A_i)\end{aligned}$$



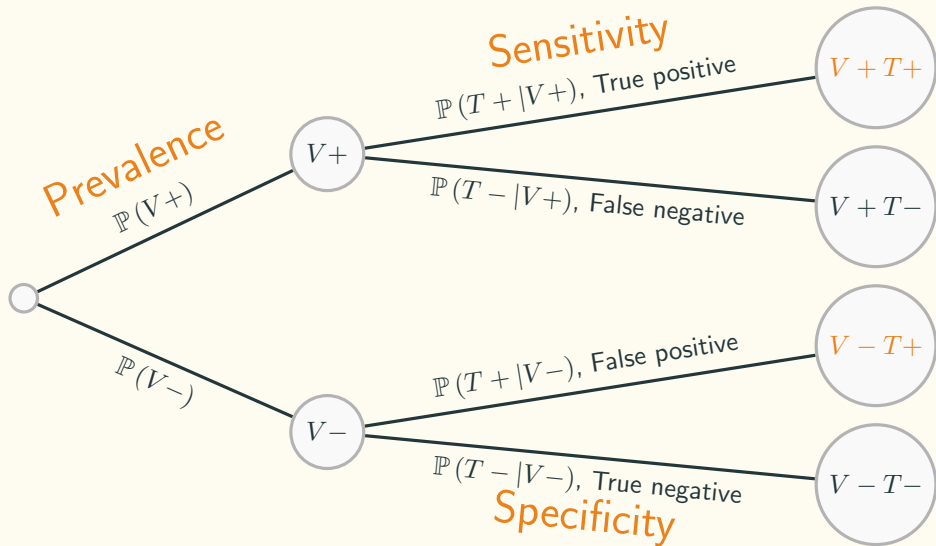
Conditional Probability

Bayes' Theorem

$$\begin{aligned}\mathbb{P}(A_i|B) &= \frac{\mathbb{P}(A_i) \cdot \mathbb{P}(B|A_i)}{\mathbb{P}(B)} \\ &= \frac{\mathbb{P}(A_i) \cdot \mathbb{P}(B|A_i)}{\sum_{i=1}^n \mathbb{P}(A_i) \cdot \mathbb{P}(B|A_i)}\end{aligned}$$



Virus Detection



ARTICLES

Judgment under Uncertainty: Heuristics and Biases

Amos Tversky¹, Daniel Kahneman¹

¹Hebrew University, Jerusalem, Israel

– Hide authors and affiliations

Science 27 Sep 1974;
Vol. 185, Issue 4157, pp. 1124-1131
DOI: 10.1126/science.185.4157.1124

Article

Info & Metrics

eLetters

 PDF

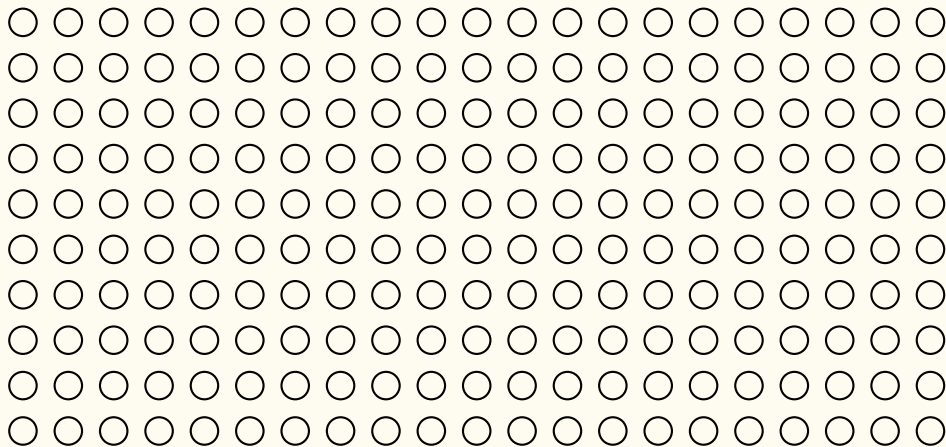
Abstract

This article described three heuristics that are employed in making judgements under uncertainty: (i) representativeness, which is usually employed when people are asked to judge the probability that an object or event A belongs to class or process B; (ii) availability of instances or scenarios, which is often employed when people are asked to assess the frequency of a class or the plausibility of a particular development; and (iii) adjustment from an anchor, which is usually employed in numerical prediction when a relevant value is available. These heuristics are highly economical and usually effective, but they lead to systematic and predictable errors. A better understanding of these heuristics and of the biases to which they lead could improve judgements and decisions in situations of uncertainty.

Amos Tversky & Daniel Kahneman

“Steve is very **shy and withdrawn**, invariably helpful, but with little interest in people, or in the world of reality. A **meek and tidy soul**, he has a need for **order and structure**, and a **passion for detail**. How do people assess the probability that Steve is engaged in a particular occupation from a list of possibilities (for example, farmer, salesman, airline pilot, librarian, or physician)?”

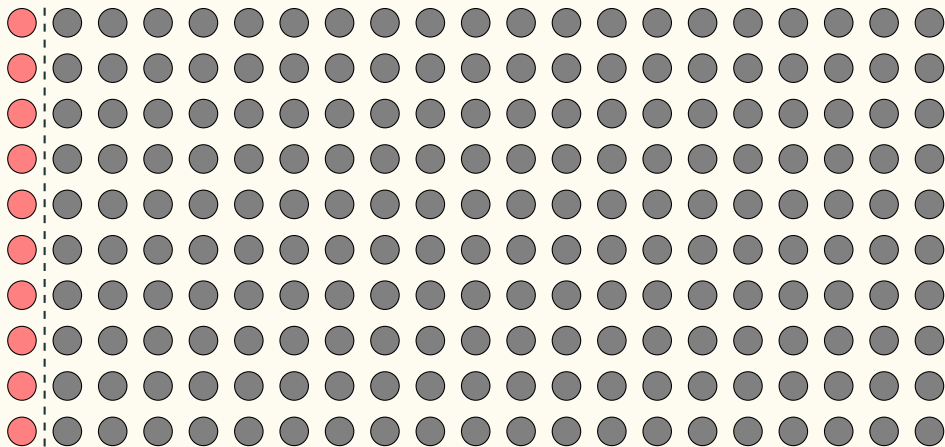
Who Is Steve



Librarian

Farmer

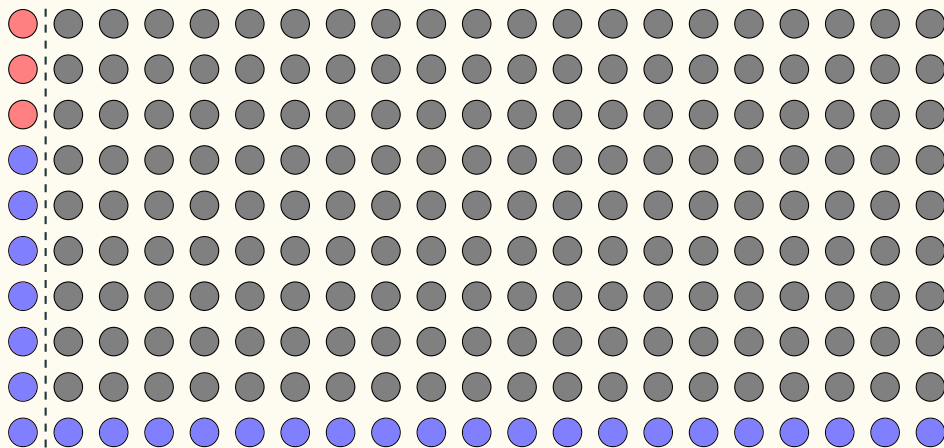
Who Is Steve



Librarian

Farmer

Who Is Steve



Librarian

Farmer

When To Use The Bayes' Theorem

You have a
hypothesis

You have observed some **evidence**

You want

The person
carries the
virus; Steve
is a librarian

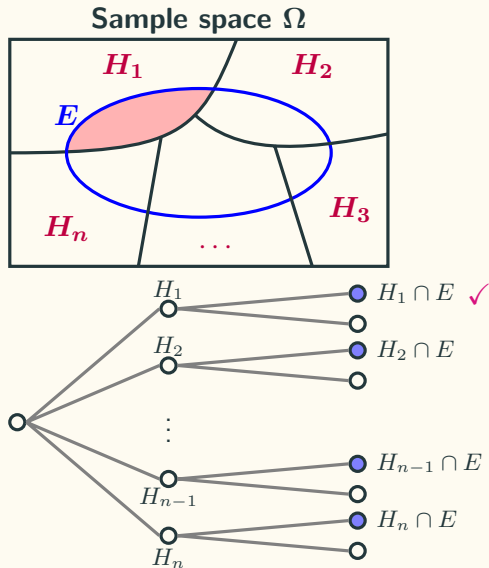
Test result is positive; Steve's characters

Probability of the
hypothesis given
the **evidence**,
 $\mathbb{P}(H|E)$

The Alternative Form Of The Bayes' Theorem

Bayes' Theorem

$$\begin{aligned}\mathbb{P}(H_i|E) &= \frac{\mathbb{P}(H_i) \cdot \mathbb{P}(E|H_i)}{\mathbb{P}(E)} \\ &= \frac{\mathbb{P}(H_i) \cdot \mathbb{P}(E|H_i)}{\sum_{i=1}^n \mathbb{P}(H_i) \cdot \mathbb{P}(E|H_i)}\end{aligned}$$



The Bayes' Theorem

$$\mathbb{P}(H_i|E) = \frac{\mathbb{P}(E|H_i)}{\sum_{i=1}^n \mathbb{P}(H_i) \cdot \mathbb{P}(E|H_i)} \cdot \mathbb{P}(H_i)$$

$\mathbb{P}(H_i)$: prior probability

$\mathbb{P}(H_i|E)$: posterior probability

The Bayesian Search

- The 4th H-bomb from American B-52 (1966)
- Air France 447 (2009 - 2011)
- Malaysian Air Flight 370 (2014 -)
- **USS Scorpion (SSN-589) (1968)**

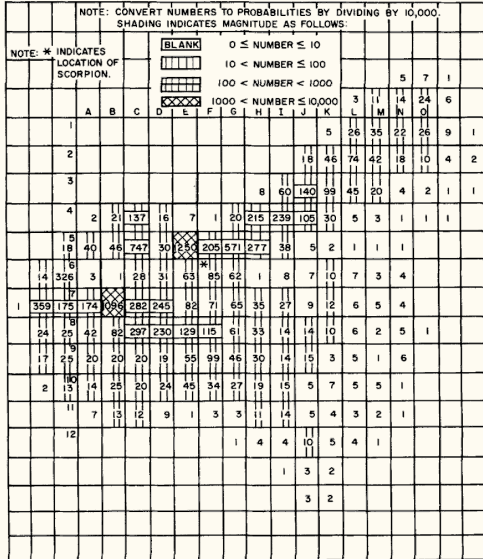
The Bayesian Search

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US Navy photo #NH_97214 & 1136658

The Bayesian Search



Richardson & Stone - *Operations analysis during the underwater search for Scorpions* (1971)

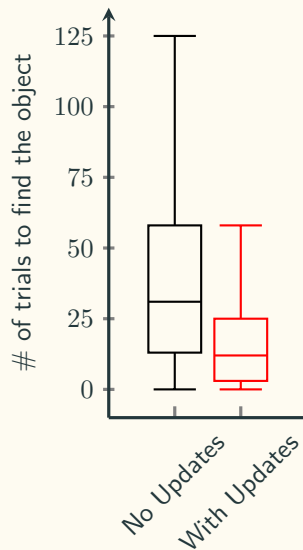
FIGURE 2. Overall A Priori distribution for *Scorpion* search

Simulation of The Bayesian Search

0.14	0.07	0.11
0.22	0.00	0.03
0.17	0.21	0.04

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One Simulation Result

0.14	0.07	0.11
0.22	0.00	0.03
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One Simulation Result

0.14	0.07	0.11
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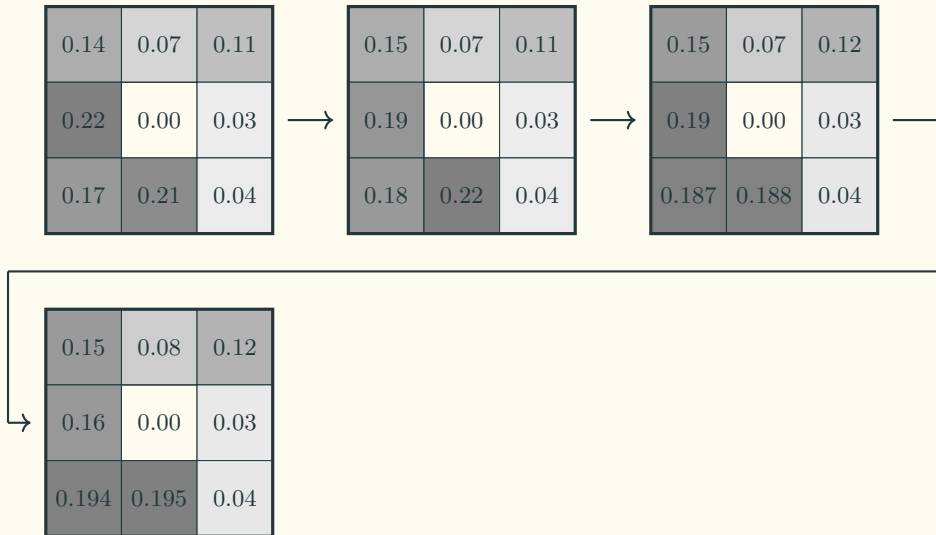
→

0.15	0.07	0.11
0.19	0.00	0.03
0.18	0.22	0.04

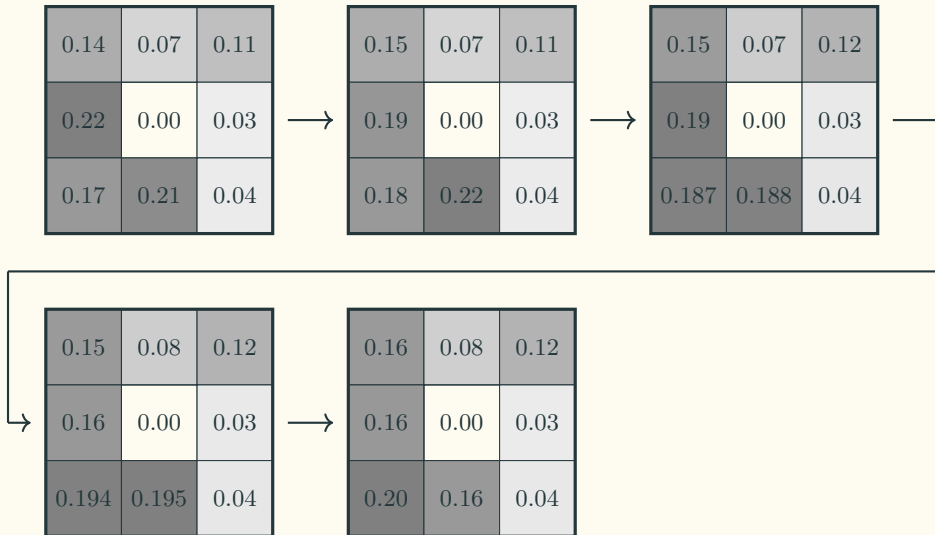
One Simulation Result



One Simulation Result



One Simulation Result



One Simulation Result

